CSC 6033 Final Exam Fall 1 2025

You have four hours to solve all eight questions.

Submit your answers in a single .pdf file.

1) Basic C++ programming language commands

Write a C++ program that reads two Integers from the standard input and compute their GCD (Greatest Common Divisor). The GCD among two numbers is the greatest Integer that is the greatest common factor number that divides them, exactly. Any correct implementation is fine.

One simple implementation is the Euclid recursive implementation that computes the GCD of two Integers ***a*** and ***b*** with a recursive function ***gcd(a, b)*** that delivers ***a***, if ***b*** is equal to zero, or delivers the ***gcd(b, c)*** where ***c*** is the remainder of the Integer division of ***a*** by ***b*** (***c = a*** mod ***b***).

For example:

- the GCD between 12 and 8 is computed as:
  - gcd(12, 8) = gcd(8, 12 mod 8) = gcd(8, 4)
  - gcd(8, 4) = gcd(4, 8 mod 4) = gcd(4, 0)
  - gcd(4, 0) = 4
- the GCD between 48 and 18 is computed as:
  - gcd(48, 18) = gcd(18, 48 mod 18) = gcd(18, 12)
  - gcd(18, 12) = gcd(12, 18 mod 12) = gcd(12, 6)
  - gcd(12, 6) = gcd(6, 12 mod 6) = gcd(6, 0)
  - gcd(6, 0) = 6

Run the code and and print out the execution for this three sets of numbers as test case:

- 513 and 39
- 1812 and 210
- 1533 and 133

Deliverable

- The code including the main function calling the three test cases;
- The output of the program generated for the three test cases.

2) **C++ programming language development**

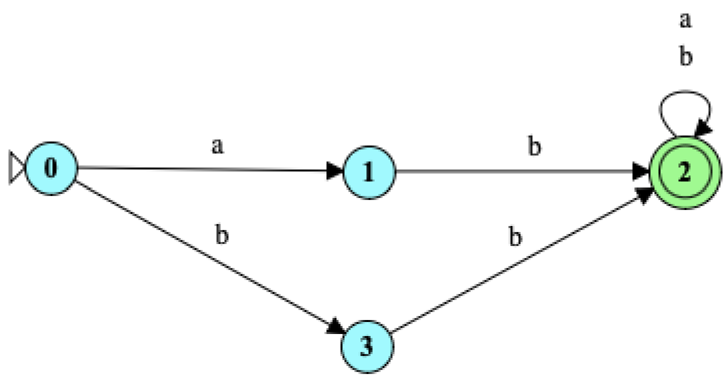
Considering the C++ code below, answer the following items.

```cpp
1  #include <iostream>
2  
3  class Full_Date {
4      public:
5          int m = 1;
6          int d = 1;
7          int y = 1;
8  
9          Full_Date() {
10             y = 2000;
11         }
12         Full_Date(int mon, int day) {
13             m = mon;
14             d = day;
15             y = 2020;
16         }
17         int getM() const  { return m; }
18         int getD() const  { return d; }
19         int getY() const  { return y; }
20         void setY(int yr) { y = yr; }
21 };
22 
23 int main()
24 {
25     Full_Date d1(3,14);
26     Full_Date d2;
27     d1.setY(2012);
28     std::cout << d1.getY() << " and " << d2.getY() << std::endl;
29     return 0;
30 }
```

A. What will be printed executing the code above?
B. What is the simplest way to set a variable of the class ***Full_Date*** to January 26 2020?
C. Are there any empty constructors in this class Full_Date?
    a. If there is(are) in which code line(s)?
    b. If there is not, how would an empty constructor be? (create the code lines for it)
D. Can the command ***std::cout << d1.m << std::endl;*** be included after line 28 without causing an error?
    a. If it can, what will be printed?
    b. If it cannot, how could this command be fixed?

3) **Finite State Automata**

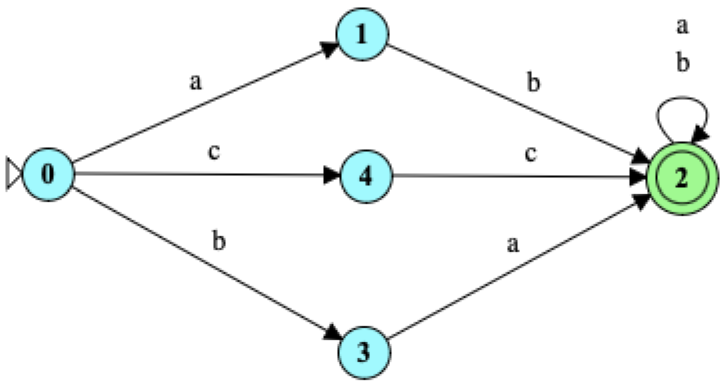
Given the FSA below, answer the following questions.



A. Give three 4-letter words that can be accepted by this FSA;
B. Give three 4-letter words that cannot be accepted by this FSA;
C. How could you describe the words accepted by this FSA?
D. Is this FSA deterministic or non-deterministic?

4) **Regular Expressions**

Given the FSA below, answer the following questions.



A. Write a RegEx that is equivalent to this FSA as it is;
B. Write a RegEx that is equivalent to this FSA removing the states and edges corresponding to the letter ***c***.

Note: To both items feel free to use any method you want, including analyzing which words are accepted by the FSA, to generate your RegEx.

5) **Pushdown Automata and Context-Free Grammars**

Considering the CFG described below, answer the following questions.

- ***Σ = {a, b}***
- ***NT = {S}***

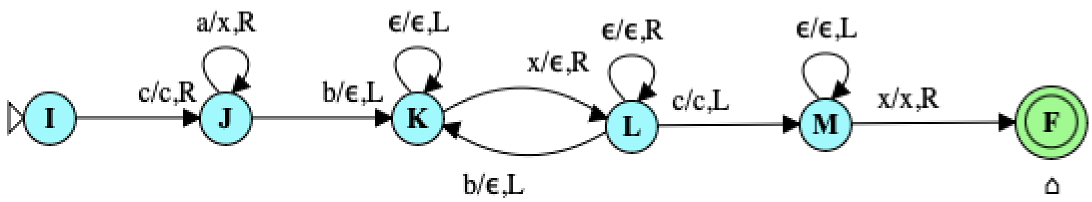
Productions:

P1	***S ⇒ aSa***
P2	***S ⇒ bSb***
P3	***S ⇒ a***
P4	***S ⇒ b***

A. List one sequence of productions that can accept the word ***abaaaba***;
B. Give three 5-letter words that can be accepted by this CFG;
C. Create a Pushdown automaton capable of accepting the language accepted by this CFG.

6) Turing Machines

Considering the TM below, answer the following questions.



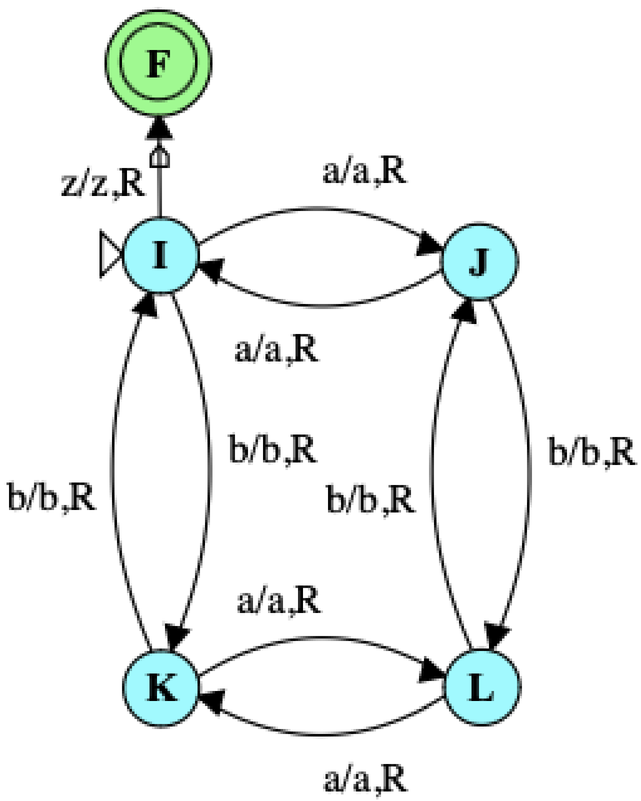
A. Give three 4-letter words that can be accepted by this TM;
B. Give three 4-letter words that cannot be accepted by this TM;
C. How could you describe the words accepted by this TM?
D. What is the alphabet of the language accepted by this TM?

7) Turing Machines with input and output

Transform the TM below that accepts words over the alphabet ***Σ = {a, b}*** with an even number of ***a***'s and ***b***'s in order that the output tape head is positioned over the first letter of the input, if the word is accepted, and all letters ***a*** should be replaced by the letter ***x***.

For example, for the input ***aabbaa*** the tape and head at the end should be:

[x]xbbxx



8) **Decidability**

Given the C++ code below, create a TM that performs the same operation, i.e., given an input over the alphabet ***Σ* = {*a*, *b*}** it prints the number of letters ***b*** in binary.

```
#include <iostream>
#include <string>

int main() {
  std::string str;
  int count = 0;
  char buffer [1000];

  std::cout << "Enter a string: ";
  std::cin >> str;
  for (char c : str) {
    if (c == 'b')
      count++;
  }
  std::string binary = "";
  while (count > 0) {
    binary = std::to_string(count % 2) + binary;
    count /= 2;
  }
  std::cout << binary << std::endl;
  return 0;
}
```